```
clc;
clear;
close all;

%% INPUT PARAMETERS
b = [0.0048 0.0193 0.0289 0.0193 0.0048];
a = [1 -2.3695 2.3140 -1.0547 0.1874];

Bc = 8;         % Coefficient word length
Bp = 16;        % Product word length
N = 200;

x = [1 zeros(1,N-1)];    % Unit impulse
n = 0:N-1;

%% 1. COEFFICIENT QUANTIZATION
S = 2^(Bc-1)-1;

bq = round(b*S)/S;
aq = round(a*S)/S;

CoeffError = [b-bq, a-aq];

%% 2. IDEAL AND QUANTIZED RESPONSES
y_ideal = filter(b,a,x);
y_coeff = filter(bq,aq,x);

e_coeff = y_ideal-y_coeff;
MaxCoeffError = max(abs(e_coeff));

SQNR_coeff = 10*log10( ...
    sum(y_ideal.^2)/max(sum(e_coeff.^2),eps));

%% 3. PRODUCT QUANTIZATION
Sp = 2^(Bp-1)-1;
y_prod = zeros(1,N);
v = zeros(1,4);

for k = 1:N
    p = [bq*x(k), -aq(2:5).*v];

    % Quantize individual products
    p = round(p*Sp)/Sp;

    % Quantize accumulated output
    acc = sum(p);
    y_prod(k) = round(acc*Sp)/Sp;

    % Update feedback states
```

```
    v = [y_prod(k), v(1:3)];
end

e_prod = y_coeff-y_prod;
MaxProdError = max(abs(e_prod));

SQNR_prod = 10*log10( ...
    sum(y_coeff.^2)/max(sum(e_prod.^2),eps));

%% 4. ZERO-INPUT LIMIT-CYCLE TEST
x0 = zeros(1,N);
state = [0.01 0.01 0.01 0.01];
y_zero = zeros(1,N);

for k = 1:N
    acc = -aq(2:5)*state.';
    y_zero(k) = round(acc*Sp)/Sp;
    state = [y_zero(k), state(1:3)];
end

LimitAmplitude = max(abs(y_zero));
LimitSamples = find(abs(y_zero)>1e-12);

%% 5. OVERFLOW AND SATURATION TEST
xo = 1.2*ones(1,N);
y_over = filter(bq,aq,xo);

% Saturation to normalized range [-1, 1]
y_sat = min(max(y_over,-1),1);

OverflowCount = sum(abs(y_over)>1);
MaxOverflow = max(abs(y_over));

%% 6. DISPLAY RESULTS
fprintf('\n=======================================\n');
```

```
=======================================
```

```
fprintf(' IIR FILTER QUANTIZATION RESULTS\n');
```

```
 IIR FILTER QUANTIZATION RESULTS
```

```
fprintf('=======================================\n');
```

```
=======================================
```

```

fprintf('Coefficient Word Length = %d bits\n',Bc);
```

```
Coefficient Word Length = 8 bits
```

```
fprintf('Product Word Length = %d bits\n',Bp);
```

```
Product Word Length = 16 bits
```

```
fprintf('\nCoefficient Quantization:\n');
```

```
Coefficient Quantization:
```

```
fprintf('Maximum Output Error = %.6e\n',MaxCoeffError);
```

```
Maximum Output Error = 4.405764e-03
```

```
fprintf('Coefficient SQNR = %.2f dB\n',SQNR_coeff);
```

```
Coefficient SQNR = 32.23 dB
```

```
fprintf('\nProduct Quantization:\n');
```

```
Product Quantization:
```

```
fprintf('Maximum Product Error = %.6e\n',MaxProdError);
```

```
Maximum Product Error = 1.651951e-01
```

```
fprintf('Product SQNR = %.2f dB\n',SQNR_prod);
```

```
Product SQNR = 1.57 dB
```

```
fprintf('\nZero-Input Test:\n');
```

```
Zero-Input Test:
```

```
fprintf('Maximum Amplitude = %.6e\n',LimitAmplitude);
```

```
Maximum Amplitude = 9.216590e-03
```

```
fprintf('Nonzero Samples = %d\n',numel(LimitSamples));
```

```
Nonzero Samples = 177
```

```
fprintf('\nOverflow Test:\n');
```

```
Overflow Test:
```

```
fprintf('Overflow Samples = %d\n',OverflowCount);
```

```
Overflow Samples = 194
```

```
fprintf('Maximum Absolute Output = %.6e\n',MaxOverflow);
```

```
Maximum Absolute Output = 1.359910e+00
```

```
%% 7. RESULTS TABLE
Test = {
    'Coefficient Quantization'
```

```
    'Product Quantization'
    'Zero-Input Test'
    'Overflow Test'
};

Maximum_Value = [
    MaxCoeffError
    MaxProdError
    LimitAmplitude
    MaxOverflow
];

SQNR_dB = [
    SQNR_coeff
    SQNR_prod
    NaN
    NaN
];

T = table(Test,Maximum_Value,SQNR_dB);
disp(T);
```

Test	Maximum_Value	SQNR_dB
{'Coefficient Quantization'}	0.0044058	32.233
{'Product Quantization' }	0.1652	1.5673
{'Zero-Input Test' }	0.0092166	NaN
{'Overflow Test' }	1.3599	NaN

```
%% 8. DISPLAY COEFFICIENTS
fprintf('\nOriginal Numerator b:\n');
```

```
Original Numerator b:
```

```
disp(b);
```

```
    0.0048    0.0193    0.0289    0.0193    0.0048
```

```
fprintf('Quantized Numerator bq:\n');
```

```
Quantized Numerator bq:
```

```
disp(bq);
```

```
    0.0079    0.0157    0.0315    0.0157    0.0079
```

```
fprintf('Original Denominator a:\n');
```

```
Original Denominator a:
```

```
disp(a);
```

```
    1.0000   -2.3695    2.3140   -1.0547    0.1874
```

```
fprintf('Quantized Denominator aq:\n');
```

```
Quantized Denominator aq:
```

```
disp(aq);
```

```
    1.0000   -2.3701    2.3150   -1.0551    0.1890
```

```
fprintf('Coefficient Quantization Errors:\n');
```

```
Coefficient Quantization Errors:
```

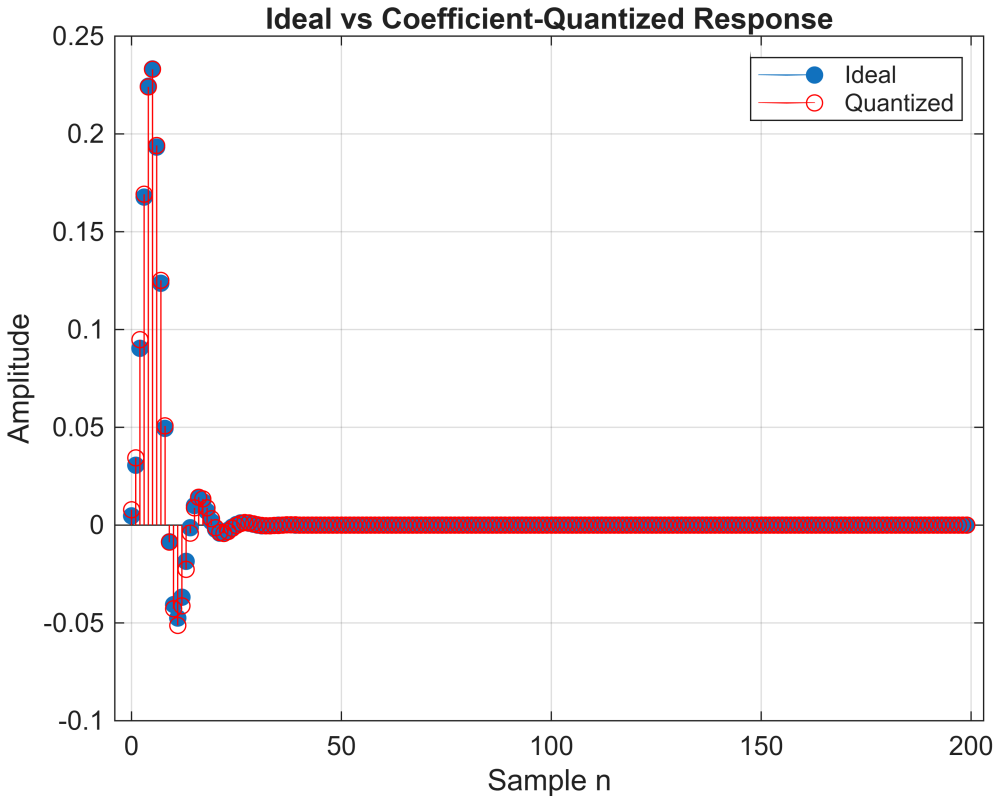
```
disp(CoeffError);
```

```
   -0.0031    0.0036   -0.0026    0.0036   -0.0031         0    0.0006   -0.0010    0.0004   -0.0016
```

```
%% 9. IDEAL VS QUANTIZED RESPONSE
figure('Name','Ideal vs Quantized Response');

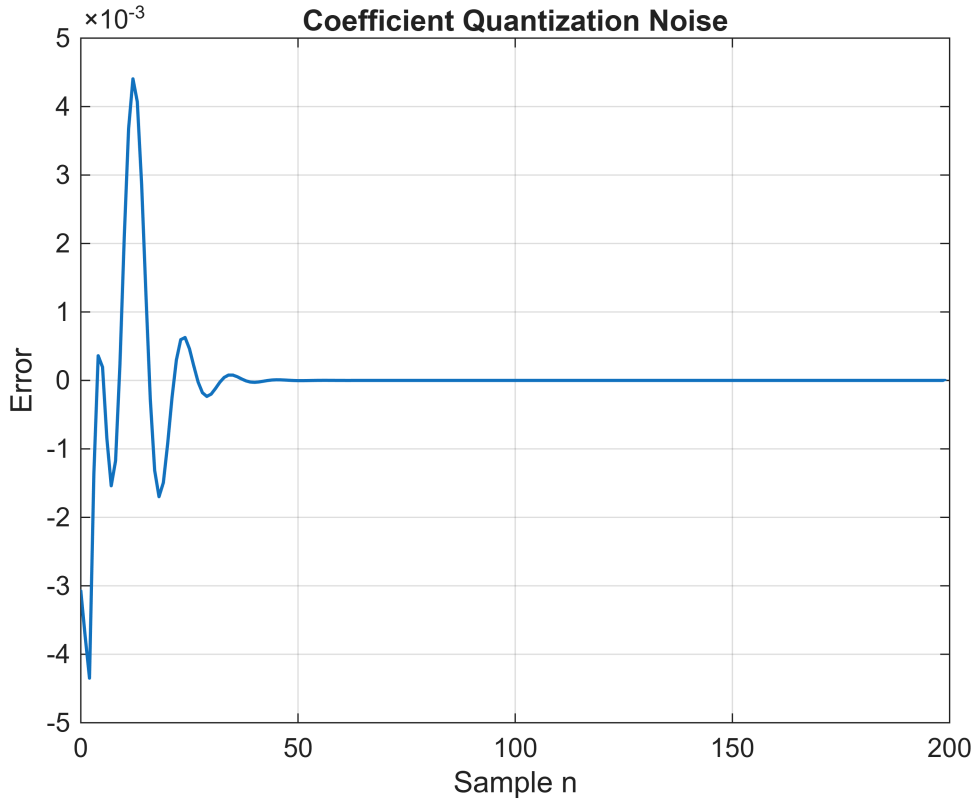
stem(n,y_ideal,'filled');
hold on;
stem(n,y_coeff,'r');

grid on;
xlabel('Sample n');
ylabel('Amplitude');
title('Ideal vs Coefficient-Quantized Response');
legend('Ideal','Quantized');
```



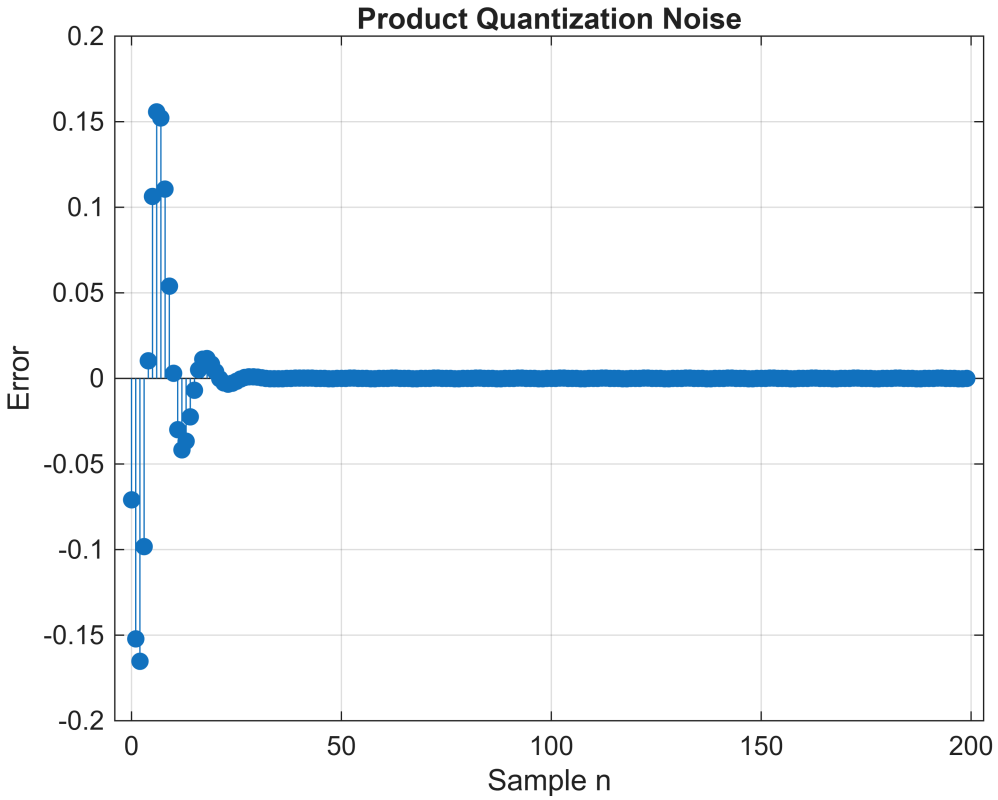

```
%% 10. COEFFICIENT QUANTIZATION ERROR
figure('Name','Coefficient Quantization Noise');

plot(n,e_coeff,'LineWidth',1.2);
grid on;
xlabel('Sample n');
ylabel('Error');
title('Coefficient Quantization Noise');
```



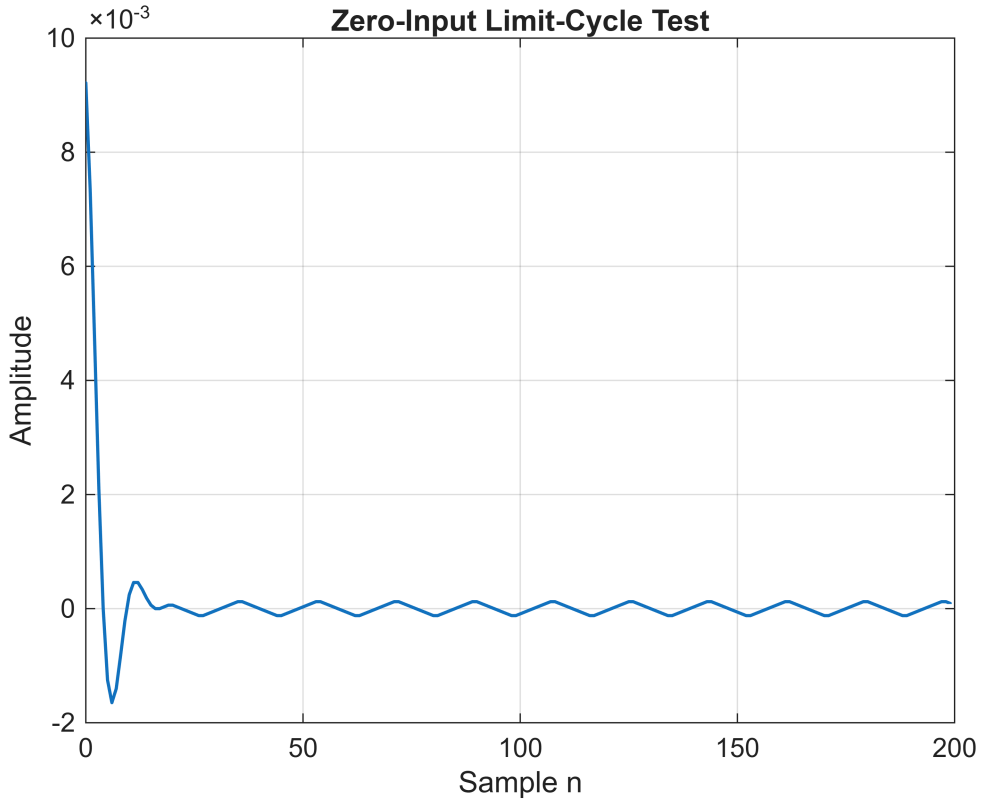

```
%% 11. PRODUCT QUANTIZATION ERROR
figure('Name','Product Quantization Noise');

stem(n,e_prod,'filled');
grid on;
xlabel('Sample n');
ylabel('Error');
title('Product Quantization Noise');
```




```
%% 12. ZERO-INPUT LIMIT-CYCLE TEST
figure('Name','Zero-Input Test');

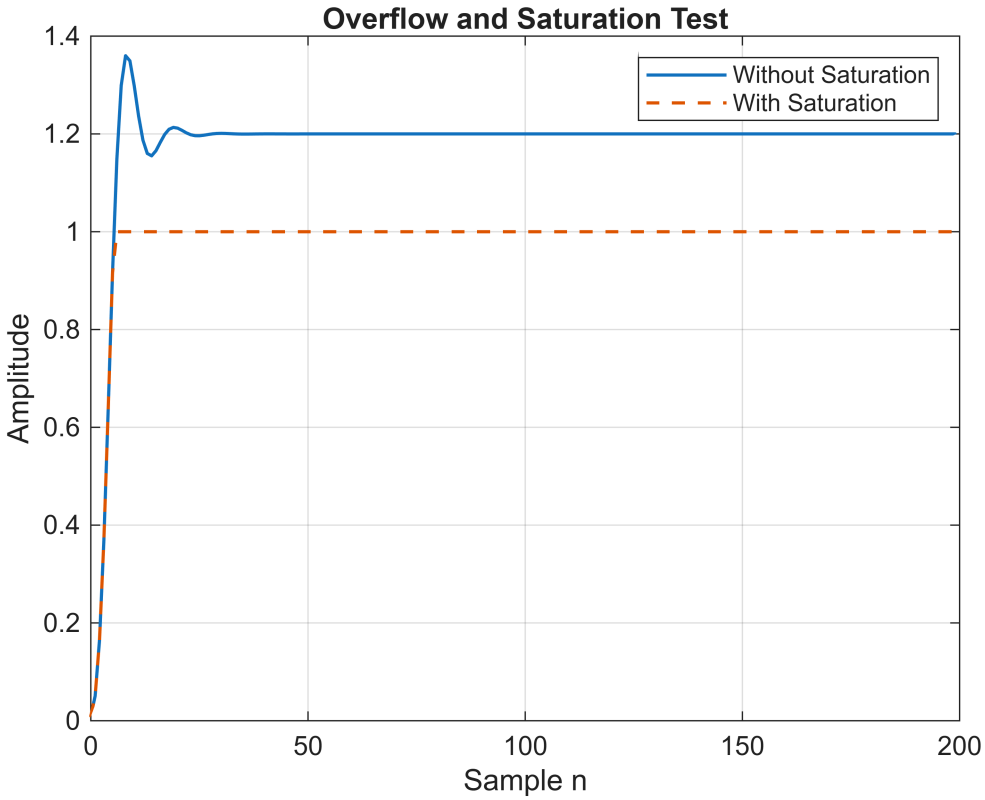
plot(n,y_zero,'LineWidth',1.2);
grid on;
xlabel('Sample n');
ylabel('Amplitude');
title('Zero-Input Limit-Cycle Test');
```




```
%% 13. OVERFLOW AND SATURATION
figure('Name','Overflow and Saturation');

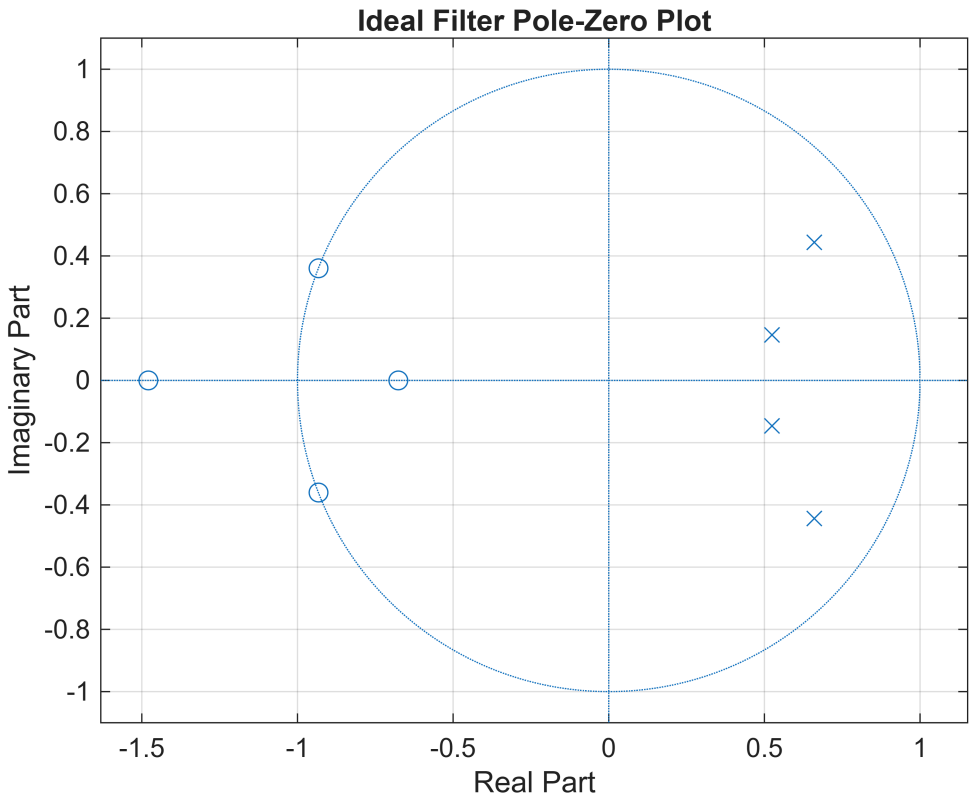
plot(n,y_over,'LineWidth',1.2);
hold on;
plot(n,y_sat,'--','LineWidth',1.2);

grid on;
xlabel('Sample n');
ylabel('Amplitude');
title('Overflow and Saturation Test');
legend('Without Saturation','With Saturation');
```



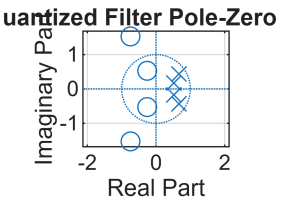

```
%% 14. IDEAL FILTER POLE-ZERO PLOT
figure('Name','Ideal Filter Pole-Zero Plot');

zplane(b,a);
grid on;
title('Ideal Filter Pole-Zero Plot');
```




```
%% 15. QUANTIZED FILTER POLE-ZERO PLOT
figure('Name','Quantized Filter Pole-Zero Plot');

zplane(bq,aq);
grid on;
title('Quantized Filter Pole-Zero Plot');
```




```
%% 16. FINAL RESULT
fprintf('\n=======================================\n');
```

```
=======================================
```

```
fprintf('FINAL RESULT\n');
```

```
FINAL RESULT
```

```
fprintf('=======================================\n');
```

```
=======================================
```

```
fprintf('1. Coefficient quantization completed.\n');
```

```
1. Coefficient quantization completed.
```

```
fprintf('2. Product quantization simulated.\n');
```

```
2. Product quantization simulated.
```

```
fprintf('3. Zero-input response evaluated.\n');
```

```
3. Zero-input response evaluated.
```

```
fprintf('4. Overflow and saturation evaluated.\n');
```

```
4. Overflow and saturation evaluated.
```

```
fprintf('5. Pole-zero plots compared.\n');
```

```
5. Pole-zero plots compared.
```

```

```